

THE REPRESENTATION UTILITY GAP IN AUDIO LLMs FOR CLINICAL SPEECH

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ABSTRACT

Audio language models answer clinical questions about a recording directly: is this speaker depressed, cognitively impaired, or has Parkinson’s? We test whether those answers use what the models hold. On Parkinson’s and Alzheimer’s, linear probes read the disease from the frozen encoder of an audio LLM at 0.78 to 0.96 AUC while the same model’s answers on the same clips sit at 0.54 to 0.66; on depression probe and answer both sit near 0.66, and the evidence is a conflict test. The answer follows the words: transcript-only input scores the same or better, synthetic speech carrying the same words scores the same as the patient’s voice, and when the words contradict the label six of nine models fall to 0.36 or lower. Seven fixes that leave the weights untouched do not close the gap; training the smallest parts closes most of it, a linear readout reaching 0.78 and a retrained projector 0.80 where the answer reads 0.62. The gap replicates on a second Alzheimer’s dataset and on pooled ADRess-2020. The diagnosis is in the model’s states; the step that maps them to an answer does not use it.

Index Terms— audio language models, clinical speech, probing, Alzheimer’s disease, Parkinson’s disease, depression

1. INTRODUCTION

Speech carries measurable signs of Parkinson’s disease, depression and Alzheimer’s disease, and audio language models (audio LLMs) now answer clinical questions about a recording in natural language. That makes them attractive for screening: one model, one prompt, no task-specific training. Probing studies add to the appeal by showing that speaker state is richly encoded across the layers of these models [1].

The unexamined assumption is that the answer reflects what the layers hold. Clinical evaluations score answers on data where the spoken words and the diagnosis point the same way, so a model that reads the transcript and a model that listens to the voice cannot be told apart on that data. Two recent studies find heavy reliance on text for emotion and synthetic paralinguistic tasks [2, 3]; multi-dataset clinical benchmarks exist but evaluate no audio LLMs, or evaluate text-only models on transcripts [4, 5]. None compares, inside one model, the information present in its representations with the information used by its answer.

We make that comparison on three conditions chosen so that how much the words should matter differs between them. In Parkinson’s read speech every speaker reads the same passage, so the words carry almost nothing. In depression interviews the words carry part of the label. In Alzheimer’s picture descriptions the words carry most of it. On every

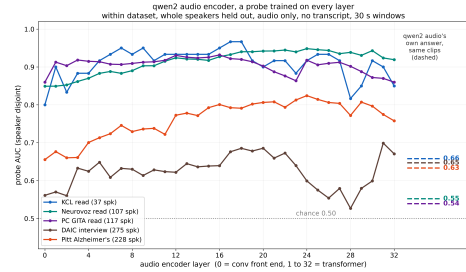


Fig. 1. A linear probe on every encoder layer of qwen2-audio (solid) against the model’s own answer on the same clips (dashed, one line for each task). On the three Parkinson’s sets the probe sits at 0.89 to 0.96 while the answers sit at 0.54 to 0.66; on Alzheimer’s the probe reaches 0.78 under strict speaker-disjoint selection while the answer reads 0.62 (the dashed Pitt line is drawn at the voice wording, 0.63). On depression at a 30s window both are near 0.66.

condition we train a linear probe on the frozen model’s hidden states and query the same model, on the same clips, with the same question.

Three findings follow. (i) The clinical signal is present: probes read it from the encoder at 0.78 to 0.96 AUC on Parkinson’s and Alzheimer’s, and on Alzheimer’s it keeps rising through the language model’s own layers. (ii) The answer does not use it: the model’s answers sit at 0.54 to 0.66 and track the words, and when the words contradict the label six of nine models fall to 0.36 or lower on depression and qwen2-audio falls to 0.41 on Alzheimer’s. (iii) The gap is a readout problem, not a representation problem: seven fixes that leave the weights untouched fail, and the two that train a small component recover most of it on Alzheimer’s, with the projector fix moving in the same direction on Parkinson’s.

2. SETUP

Models. Nine audio LLMs, all frozen: qwen2-audio [6], qwen2.5-omni [7], qwen3-omni [8], audio flamingo 2 and 3 [9, 10], kimi-audio [11], salmonn [12], mimo-audio [13] and diva [14]. All nine answer the conflict test; the layer-wise probing and the fixes use qwen2-audio, whose encoder (33 hidden-state stages) and language model (33 stages at the audio-token positions) we probe end to end.

Datasets. Parkinson’s read speech: PC-GITA (Spanish, 1117 clips) [15], Neurovoz (Spanish, 1270) [16], MDVR-KCL (English, 37 speakers) [17]. Depression: DAIC-WOZ interviews (275 participants; the label is the dataset’s PHQ-8 self-report binary, not a clinical diagnosis) [18]. Alzheimer’s: the De-

mentiaBank Pitt cookie-theft descriptions (468 thirty-second segments, 228 speakers) [19, 20], the ADReSSo recordings (237 speakers) [21], and the ADReSS-2020 benchmark split (108 train, 48 test) [22]; on both, the 30 s window starts at the patient’s first words, not the interviewer’s prompt. An Italian Parkinson’s dataset was excluded after recording properties alone separated its groups at 0.995 (Section 2.1).

Prompt and metric. Two wordings, both verbatim, with no transcript supplied in any audio condition. On the Parkinson’s sets and the 30 s and 300 s depression answers: “Based only on how this person’s voice sounds, does this speaker show signs of X? Answer with one word, Yes or No.” On the Alzheimer’s sets, the depression conflict test and every fix: “Based only on this recording, does this speaker show signs of X? Answer with one word, Yes or No.” On Pitt the two wordings score 0.62 and 0.63; on the depression conflict pairs qwen2-audio reads 0.35 and 0.31. We read $p_{\text{yes}} = P(\text{Yes})/(P(\text{Yes}) + P(\text{No}))$ from the logits at the first answer position, one forward pass, no generation, and report AUC against the clinical label. We also record the mass $P(\text{Yes}) + P(\text{No})$ and the yes-rate, because some models answer Yes on every clip (qwen2.5-audio on PC-GITA) or No on every clip (qwen2.5-omni on Pitt and the Parkinson’s sets); those AUCs rank confidences that never cross the threshold, and we flag them.

Probe. An ℓ_2 -regularised logistic probe on the mean-pooled hidden state of one layer, with standardisation fitted inside the training fold. Folds are speaker-disjoint (grouped 5-fold). Where a layer is chosen, it is chosen inside the training folds; selecting the layer after seeing the test fold inflates AUC by +0.03 on average across 18 cells and by up to +0.14, so we quote the nested value wherever both exist.

Conflict sets. From the raw DAIC transcripts we take 26,083 turns, 23,351 of them from participants once the interviewer is removed, score each participant turn with a text sentiment model, and keep the moments where a depressed participant speaks positively (138) or a control speaks negatively (57), paired with matched agreement segments from the same speakers: 195 pairs over 74 speakers, with label and lexical direction independent ($\phi = 0.000$). On Pitt we score the words of each segment with a content-marker fluency model, after removing speaking rate, which alone reaches 0.92 and would make the split trivial; 146 segments land in conflict and 322 in agreement. Nothing is synthesised: the voices are real patients and the contradicting words occur naturally.

Transcript and synthesis controls. For the transcript-only condition the model receives the segment’s transcript (official for Pitt, with annotation codes stripped; whisper-large-v3 [23] for ADReSSo, which ships none) with the same question and no audio. For the synthesis condition each transcript is spoken by a neutral Piper voice [24] and scored like the real recording; that experiment used its own transcript preparation and text prompt.

2.1. Controls on the measurement

Six recording measurements containing no speech content separate patients from controls at 0.995 on the Italian dataset; we exclude it. On the four Parkinson’s and depression datasets the recording-only floor is 0.45 to 0.80 and the probe stays above it; on Neurovoz, age matching moves the encoder probe from 0.95 to 0.91 while a silence-only probe

falls from 0.72 to 0.58. On Pitt the six together reach 0.65, below the probe (0.78); controls are younger (64 against 72 years, age alone 0.70), but on 64 age and sex matched speaker pairs (268 segments) the probe still reads 0.79 and the answer 0.66. A classical baseline, eGeMAPS features [25] into the same logistic classifier, reads 0.62 on Pitt, 0.73 on ADReSSo, 0.70 on ADReSS-2020 and 0.80 on MDVR-KCL; the probes beat it on every set, and on Pitt the model’s own answer merely matches it.

3. RESULTS

3.1. The answers out of the box

Asked the diagnosis question directly, the answers are weak. On the Parkinson’s read-speech sets qwen2-audio’s answers read 0.54, 0.55 and 0.66, on the Pitt Alzheimer’s segments 0.62, on depression 0.65. The answers are usable where the words agree with the label: 0.70 to 0.93 for eight of nine models on the agreement segments, 0.57 for audio flamingo 2 (Table 1). They also improve with more audio: at a matched 300 s window on depression, audio flamingo 3, qwen2.5-omni and kimi-audio read 0.86, 0.84 and 0.83, where qwen2.5-omni reads 0.67 at 30 s. The audio answer beats the transcript answer only where there are few words to read, on Parkinson’s read speech, where the transcript answer sits between 0.47 and 0.67; on depression no model beats its own transcript.

3.2. Probe against answer

The encoder holds what the answer does not use (Figure 1). Under nested layer selection the encoder probe reads 0.92 on PC-GITA, 0.96 on Neurovoz and 0.89 on MDVR-KCL read speech, and 0.78 on the Pitt Alzheimer’s segments; the same model’s answers on the same clips read 0.54, 0.55, 0.66 and 0.62. On Alzheimer’s the signal does not stop at the encoder: probing the language model’s own stages at the audio-token positions, the readable signal rises to 0.86 by the deep layers while the answer stays where it was. On Alzheimer’s the words also carry the task (a text classifier reads 0.70) and the model can transcribe, so the words could be helping here too; where the words carry almost nothing, on MDVR-KCL read speech, the probe still holds 0.82 to 0.88 through every language-model stage. On depression at the 30 s window qwen2-audio accepts, the nested probe (0.66) and the answer (0.65) coincide.

3.3. What the answer uses: the words

The answer follows the words, and three controls agree. Transcript only. Given the Pitt transcript and no audio, qwen2-audio scores 0.68 (95% bootstrap interval 0.63 to 0.72); with the audio it scores 0.62 (0.56 to 0.67), and on the same clips the audio answer sits 0.06 below the transcript answer (paired interval -0.12 to $+0.01$): the audio adds nothing the words did not already carry. On ADReSSo the pair is 0.68 and 0.66. On DAIC the transcript-only answer of qwen2.5-omni correlates with its own audio answer at $r = 0.83$ and returns the identical Yes/No on 96% of segments, and across the depression grid the median audio answer (0.83) sits beside the median transcript answer (0.81). Synthesised voice. For qwen2-audio, replacing each Pitt patient’s voice with a neutral synthetic voice speaking the same words moves the

Table 1. Depression conflict test, 195 matched pairs, no transcript given. AUC against the label. Six of nine models fall to 0.36 or lower (bold) where the words point away from the label; mimo-audio reads 0.47 and kimi-audio 0.52 there; audio flamingo 2 is flat on both arms.

model	conflict	agreement
salmonn	0.29	0.85
diva	0.30	0.70
qwen2.5-omni	0.31	0.85
qwen3-omni	0.35	0.87
qwen2-audio	0.35	0.92
audio flamingo 3	0.36	0.93
mimo-audio	0.47	0.82
kimi-audio	0.52	0.87
audio flamingo 2	0.55	0.57

answer from 0.62 to 0.60, and for qwen2.5-omni from 0.68 to 0.67. On the synthetic voice alone diva reads 0.69 and mimo-audio 0.58. In that experiment’s own text prompt the transcript alone scores 0.72, 0.72 and 0.71 for the first three models, above the synthetic voice, and 0.57 for mimo-audio against 0.58 synthetic. The same shape holds on DAIC. Words check. A probe trained only on Pitt segments where the words agree with the diagnosis reaches 0.95 within that set and at most 0.58 on the segments where the words contradict it; on ADReSSo, with one clip from each speaker, the same test gives 0.99 within and 0.39 to 0.61 on conflict, on a small conflict group of 19. What the probe learns here is largely the words, and they enter at the encoder itself, which was trained to transcribe.

3.4. The stress test: words against diagnosis

Six of nine models fall to 0.36 or lower where the words point away from the label, as low as 0.29, while scoring 0.70 to 0.93 on the matched segments where words and label agree (Table 1); mimo-audio reads 0.47 and kimi-audio 0.52 on conflict. A model that listened would not invert; a model that reads must. On the Pitt Alzheimer’s conflict segments qwen2-audio falls to 0.41 (0.31 to 0.52) against 0.70 on agreement; qwen2.5-omni reads 0.44 against 0.77 on the same segments, answering No on every clip.

3.5. Fixes without training

Seven training-free fixes fail (Table 2). Twelve prompt wordings never lift the depression conflict AUC above 0.48. Learned layer mixing over the encoder reaches 0.909 against 0.916 for one layer chosen inside the training folds, and conditioning the mixture on the prompt adds 0.0004. Subtracting the transcript-only logits from the audio logits stops the model following the words at weight $\alpha = 0.9$ (agreement with the lexical direction falls from 0.70 to 0.50), where conflict AUC is 0.50; larger α reaches 0.67 only by collapsing the agreement arm to 0.18, and held-out speakers choose $\alpha = 0$ in four of five folds. Adding a multiple of the difference-in-means diagnosis direction to the encoder output damages the agreement arm at the smallest coefficient before conflict moves, and a random direction of the same norm displaces the answer equally; across 20 steered conditions the apparent gain correlates at -0.92 with the lexical signal being

Table 2. Conflict and agreement answers under each fix (AUC); the layer mixers, scored on PC-GITA overall AUC, are in the text. Training-free fixes that move conflict pay elsewhere: subtraction and steering in the agreement arm, the untrained plug in the overall answer (0.49). The omni baseline is the sweep’s rerun (0.31 in Table 1).

fix	conflict	agreement
depression, no training, qwen2.5-omni		
the model’s own answer	0.30	0.87
best of 12 prompt wordings	0.34	0.80
text subtraction, $\alpha = 0.9$	0.50	0.72
depression, no training, qwen2-audio		
the model’s own answer	0.35	0.92
best of 12 prompt wordings	0.48	0.85
activation steering, best strength	0.54	0.50
Alzheimer’s, no training, qwen2-audio		
the model’s own answer	0.41	0.70
few-shot audio examples (Yes on 1%)	0.46	0.45
best encoder layer plugged in, untrained	0.55	0.47
steering at the probe’s layer, $c = 1$	0.56	0.42
Alzheimer’s, a small part trained, qwen2-audio		
linear readout on the frozen states	0.61	0.86
projector retrained on layer 24	0.57	0.89

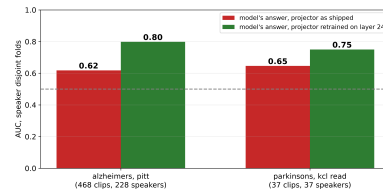


Fig. 2. Retraining only the projector, everything else frozen, speaker-disjoint folds. The model’s own answers improve on both diseases. On Alzheimer’s the gain is $+0.18$ (95% interval 0.11 to 0.24); on MDVR-KCL, with 37 speakers, it is $+0.10$ (interval -0.13 to 0.34), a direction rather than a significant result.

destroyed, and on Pitt, at the probe’s own layer 22, the diagnosis direction gives 0.56 on conflict and 0.42 on agreement against 0.58 and 0.54 for a random one. Feeding the probe’s best encoder layer into the projector without training breaks the answer (0.62 to 0.49 overall): the projector was trained on the final layer’s output and does not accept another layer’s without retraining. A checkpoint built with the VoxParadox recipe [3], prompt-conditioned layer mixing with preference tuning, reads 0.37 on the conflict pairs against 0.35 for its base model.

3.6. Fixes that train one component

Two interventions train a small component and both close most of the gap. A linear readout on the frozen hidden states, which is the probe itself, reads 0.78 (interval 0.74 to 0.83) on the Pitt segments where the answer reads 0.62, a paired gain of $+0.17$ (0.10 to 0.23). Retraining only the projector, the connector between the audio encoder and the language model, on the diagnosis answer, with the encoder and the language model frozen and encoder layer 24 as its input, lifts the model’s own answers from 0.62 to 0.80 (0.76 to 0.84). The

Table 3. A probe trained on one Parkinson’s read-speech dataset and tested cold on another (qwen2-audio encoder). Diagonal: within-dataset nested values (Section 3.2). Transfer is partial and does not follow language: Spanish to Spanish (0.46, 0.66) is worse than Spanish to English (0.86, 0.74).

trained on, tested on	KCL	Neurovoz	PC-GITA
KCL	0.89	0.58	0.64
Neurovoz	0.74	0.96	0.66
PC-GITA	0.86	0.46	0.92

paired gain is +0.18 (0.11 to 0.24), and the agreement segments move from 0.70 to 0.89. Retraining the projector on its default input, the final encoder layer, lifts the answers to 0.76 (0.72 to 0.81; agreement 0.86, conflict 0.48); the layer-24 input adds +0.035 (−0.004 to +0.073) overall and +0.089 (+0.004 to +0.177) on conflict: the training carries the gain, the earlier layer helps on conflict. On the conflict segments the readout reaches 0.61 (a paired gain of +0.19, interval 0.04 to 0.34) and the retrained projector 0.57 (+0.16, 0.02 to 0.30): both trained fixes move word conflict, and both leave it near 0.6. That the deficit sits in the last step, not earlier, is testable. A probe on the hidden state at the answer position itself, the vector from which Yes or No is read, reads 0.80 at the final layer and 0.83 at its best, and 0.69 on the conflict segments, the highest conflict number in this paper; the emitted answer on the same state reads 0.62. The language model integrates the signal; the projection to the answer tokens discards it. The same projector recipe on MDVR-KCL moves the Parkinson’s answers from 0.65, that run’s own baseline, to 0.75 (Figure 2).

3.7. Other datasets and models

On ADReSSo, 237 speakers none of whom appear in Pitt, the probe reaches 0.87 with the layer chosen inside the training folds (0.89 at the curve’s peak) and the model’s answer 0.66 (0.60 to 0.74), with transcript-only input at 0.68. On the ADReSS-2020 benchmark split, with the layer chosen inside the 108 training recordings, the probe reads the 48 test recordings at AUC 0.80 and accuracy 73%, against the challenge’s own baselines of 62.5% for acoustic features and 75% for manual transcripts [22]. The model’s answer reads 0.78 (0.63 to 0.90) on those 48 recordings but 0.62 (0.51 to 0.73) on the 108 training recordings, a swing 48 clips cannot settle, so we pool: over all 156 recordings the probe reads 0.82 with nested layer choice (0.87 at the peak) and the answer 0.68 (0.60 to 0.76). Cold cross-dataset encoder probes also exist for four further models, two outside the nine (granite speech [26] and voxtral [27]) plus qwen2.5-omni and diva; with the layer chosen on the two datasets neither trained nor tested on (the excluded Italian one among them), their best AUCs are 0.81, 0.85, 0.83 and 0.84. Across the Parkinson’s sets the probe transfers across language (Table 3).

4. DISCUSSION

Read together, the results have one shape. On Parkinson’s and Alzheimer’s the clinical signal is present at every depth of the model, the answer does not use it, and on all three

conditions what the answer uses instead is the words. Every training-free fix that moves the conflict answer pays for it elsewhere, and the best prompt wording moves qwen2-audio only from 0.35 to 0.48; the two fixes that train one small component recover most of the Alzheimer’s gap (a readout on the frozen states 0.78, the retrained projector’s own answer 0.80), and the projector fix moves the MDVR-KCL answers the same way. The deficit therefore lives in the readout, not in the representation: these models are better clinical encoders than clinical reporters, and evaluations that score their answers only where words and diagnosis agree cannot tell a model that listens from one that reads.

Limitations. The layer-wise probing and the fixes are on one model; the nine-model evidence is the answer-side conflict test. MDVR-KCL has 37 speakers and its projector gain is not significant on its own; its before number uses the voice wording, the retrained projector the recording wording. ADReSSo transcripts are ASR output (whisper-large-v3) and its conflict group has 19 segments; the ADReSS-2020 test labels were recovered by matching transcripts to Pitt (107 of 108 training labels matched, none wrong), pending the official labels. The DAIC 30s clips were not filtered for interviewer speech and will be re-cut. On DAIC, scores track PTSD severity as closely as depression severity (0.60 against 0.59). We release segment identifiers, scripts and every clip’s score.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study used existing recordings only: PC-GITA, Neurovoz, DAIC-WOZ, DementiaBank Pitt, ADReSSo and ADReSS-2020 under each distributor’s data agreement, and MDVR-KCL under its CC BY 4.0 licence. No new recordings were made and no audio is redistributed.

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